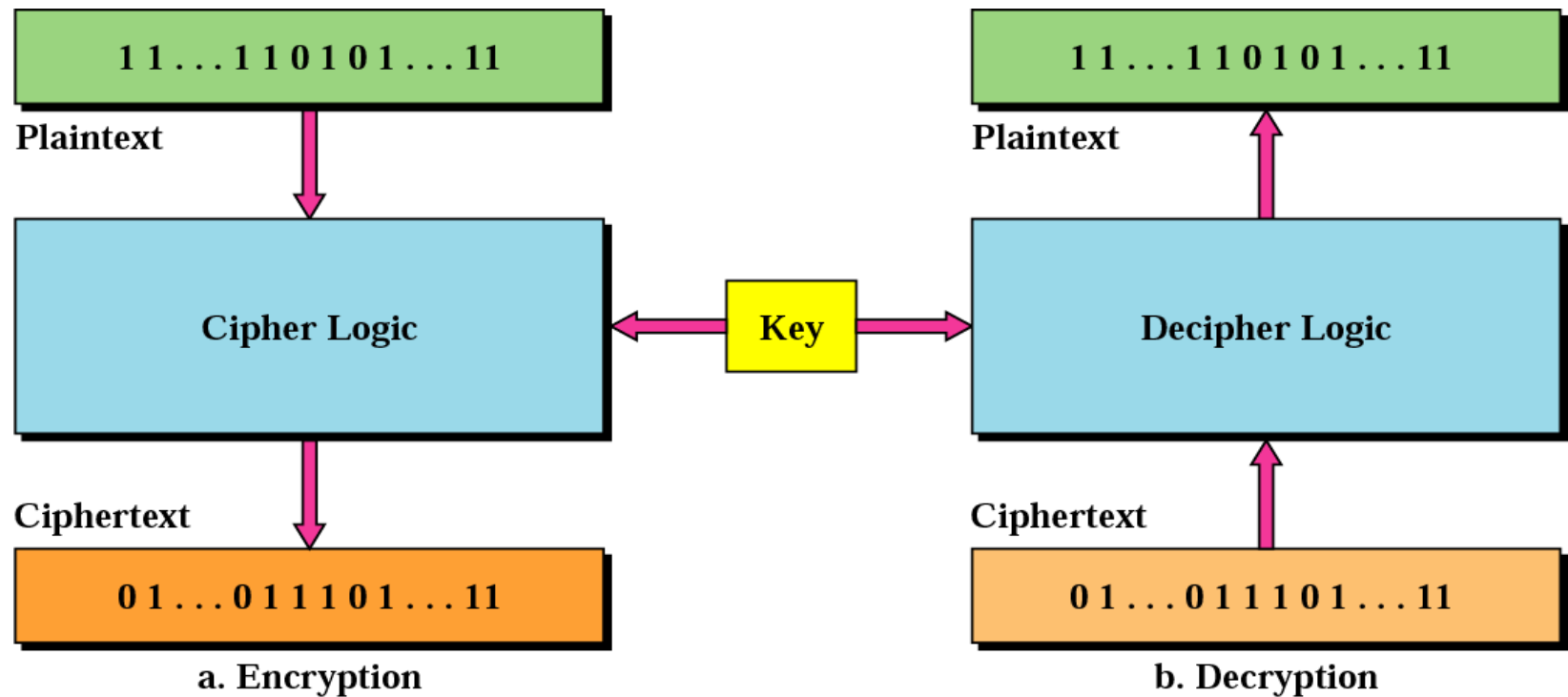
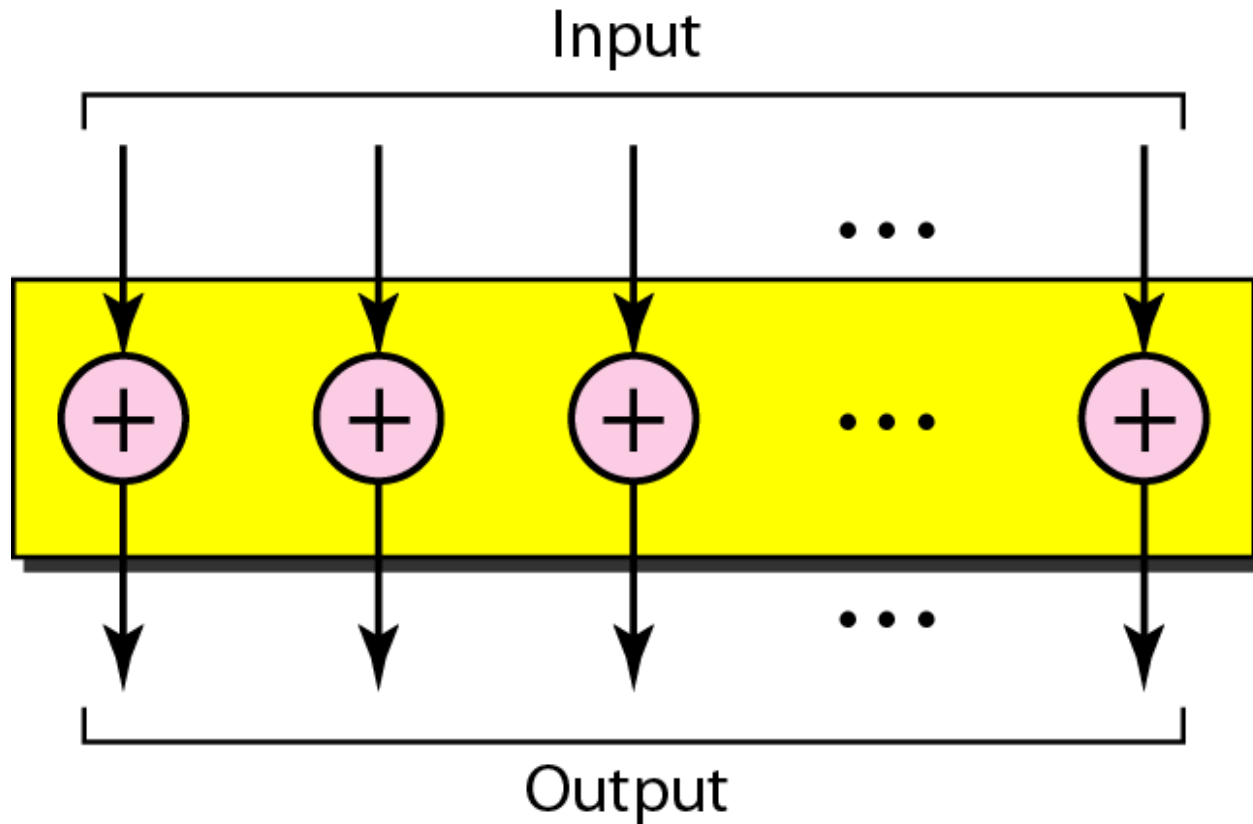


Data Encryption Standard *(DES)*

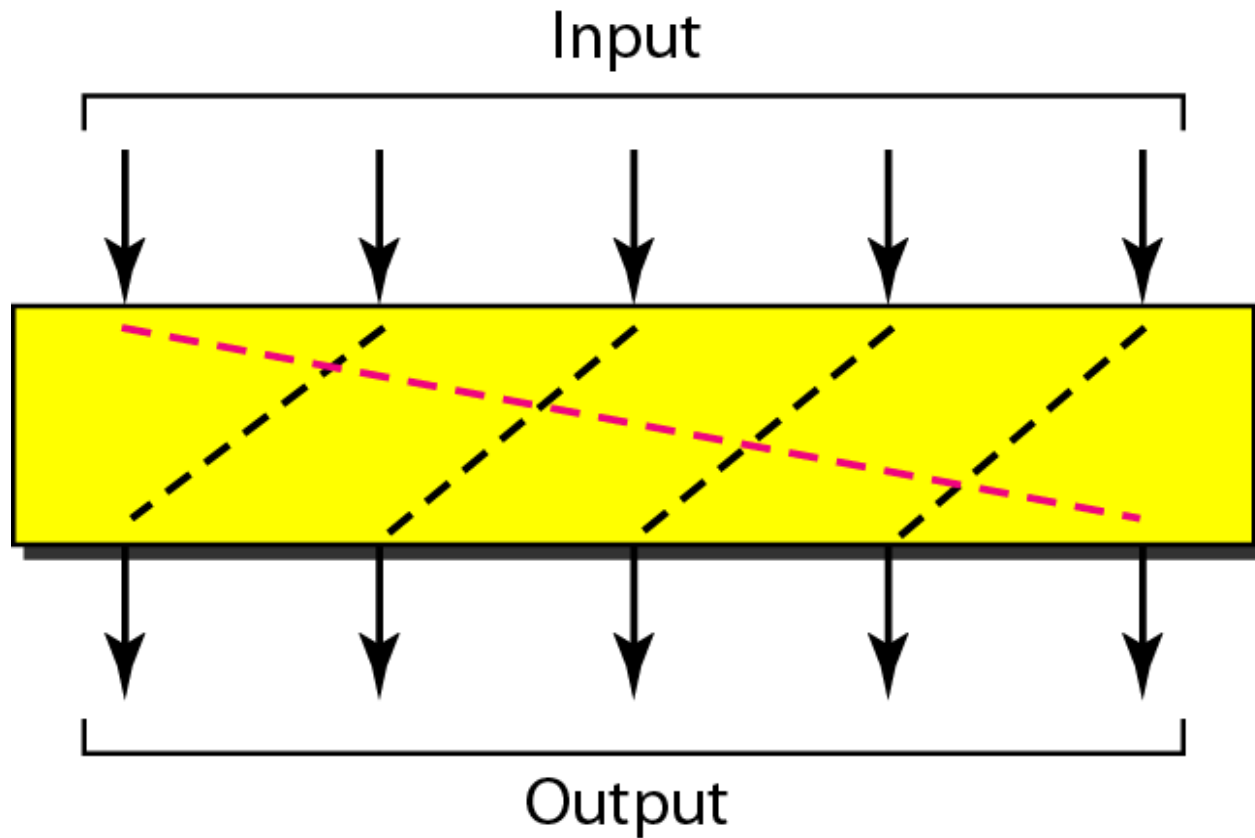
Block Cipher



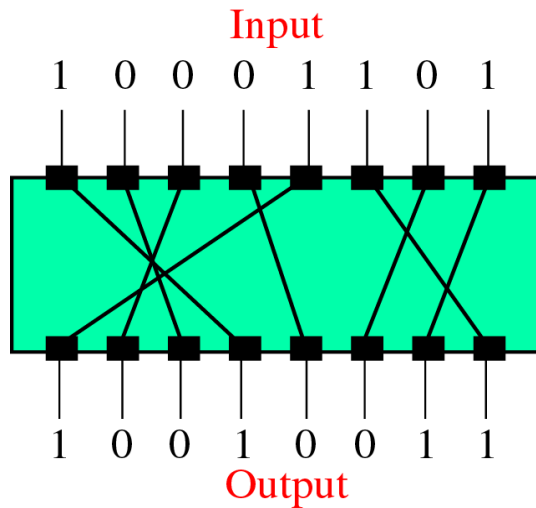
XOR Cipher



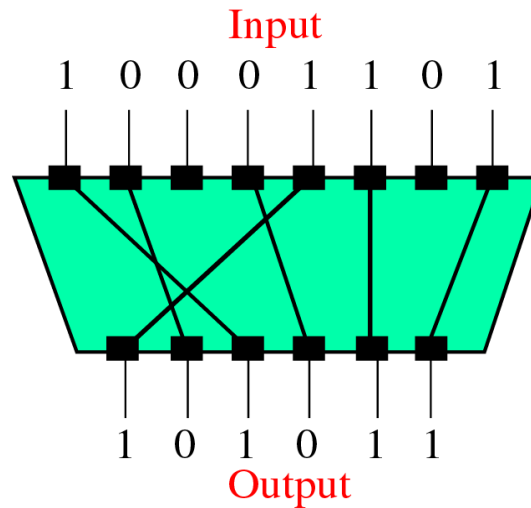
Rotation Cipher



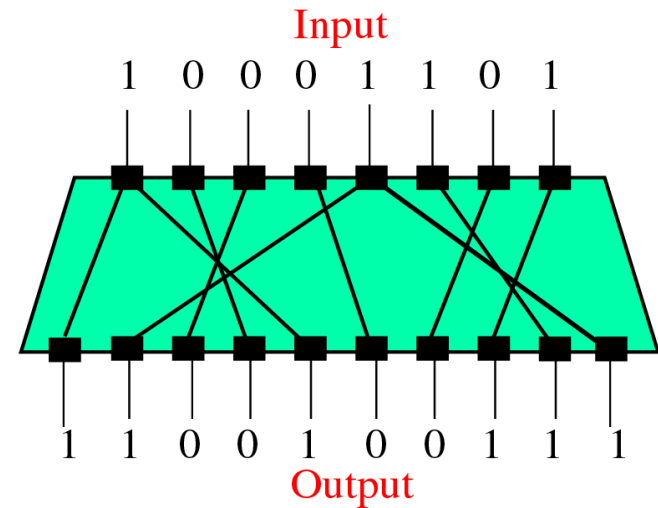
P-Box: Permutation



a. Straight permutation

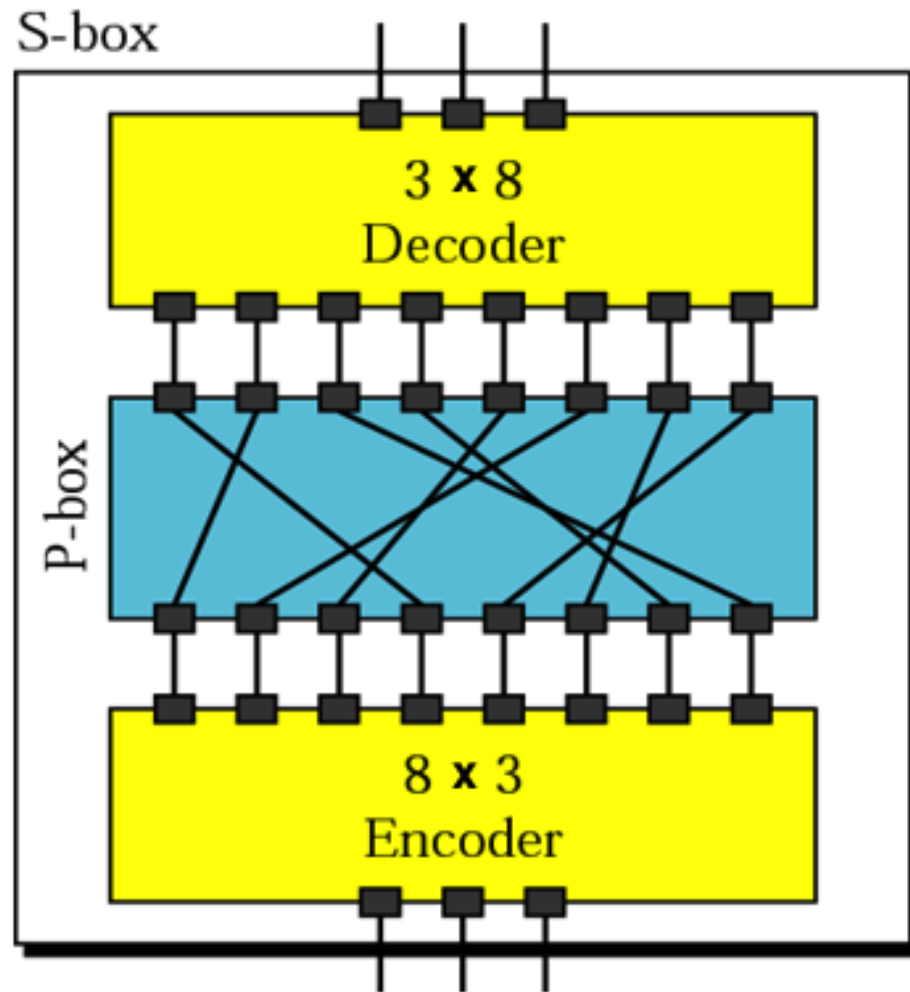


b. Compressed permutation

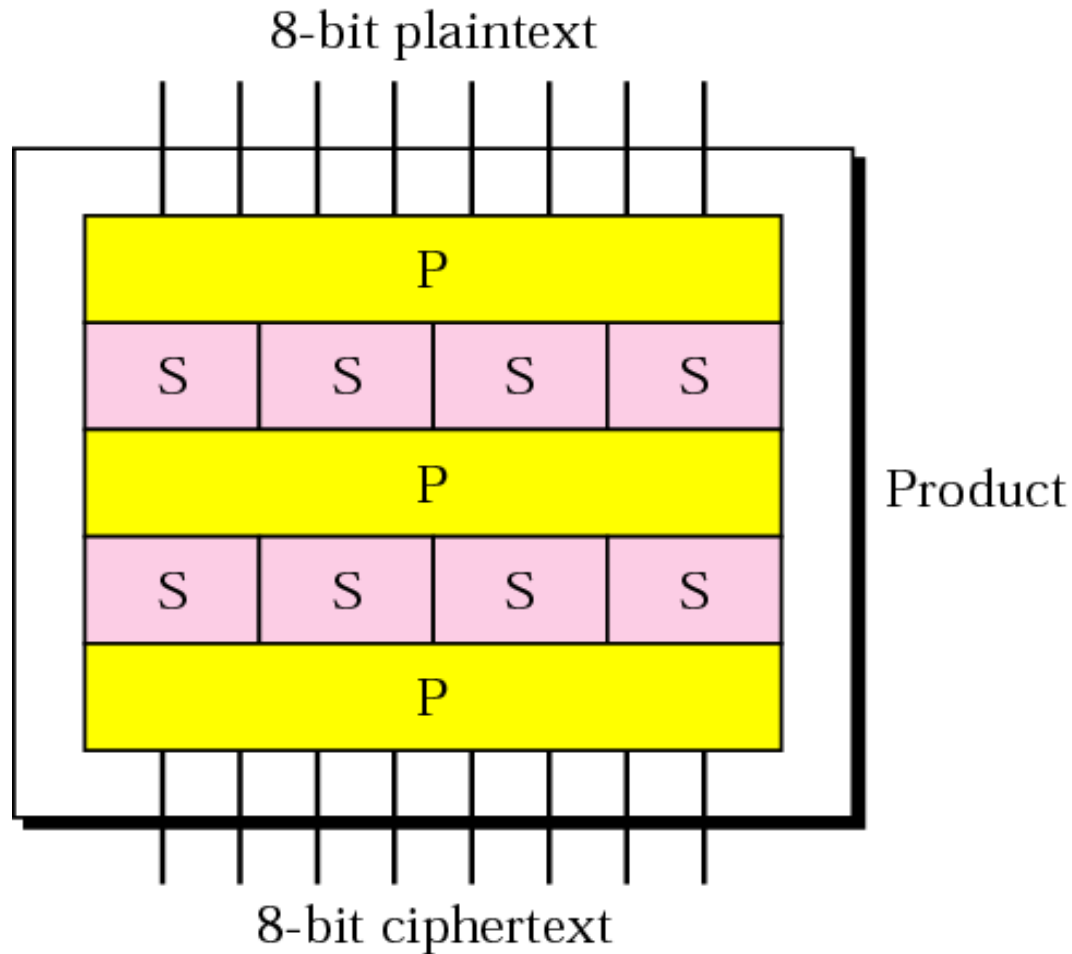


c. Expanded permutation

S-Box: Substitution



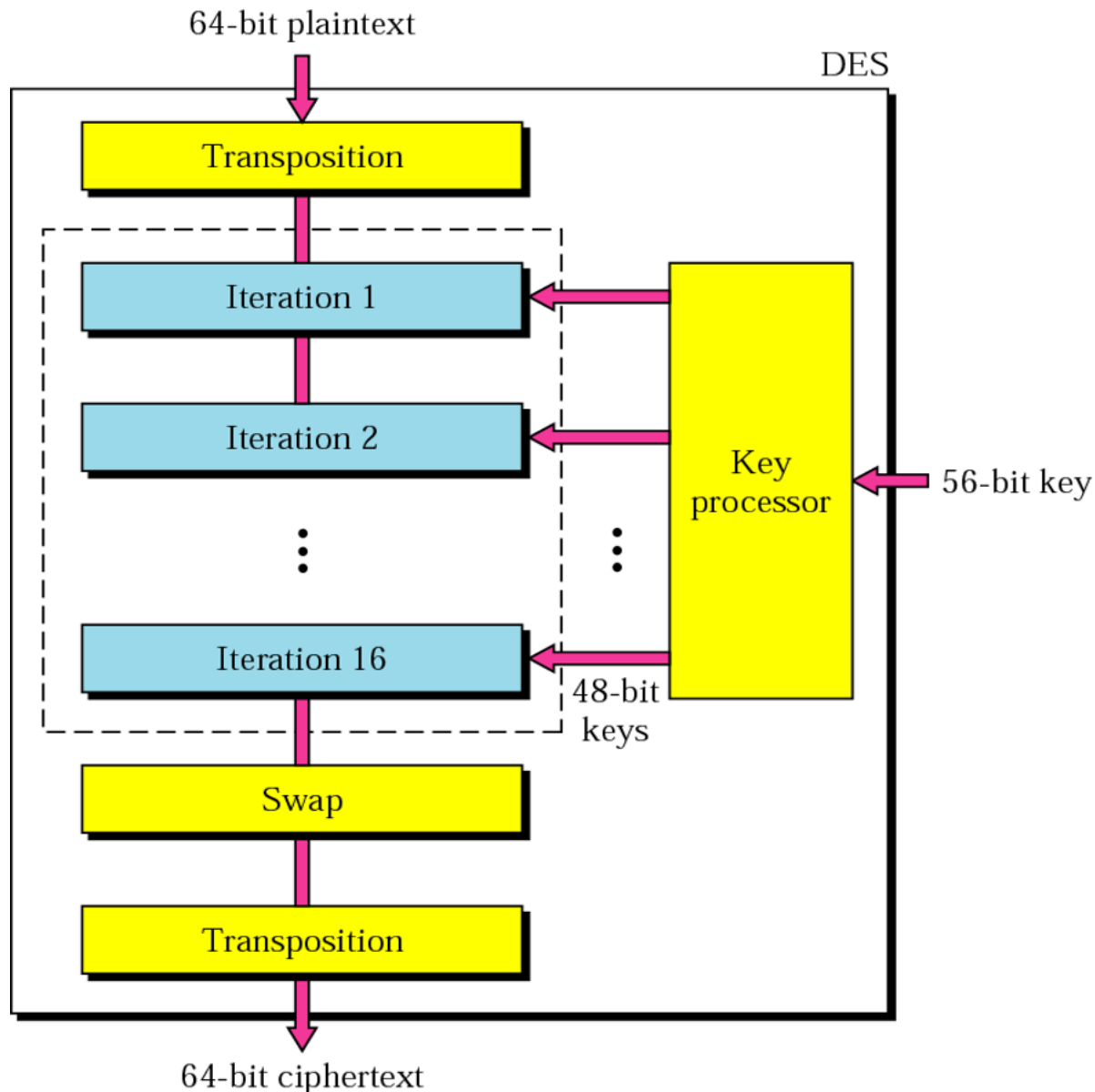
Product Block



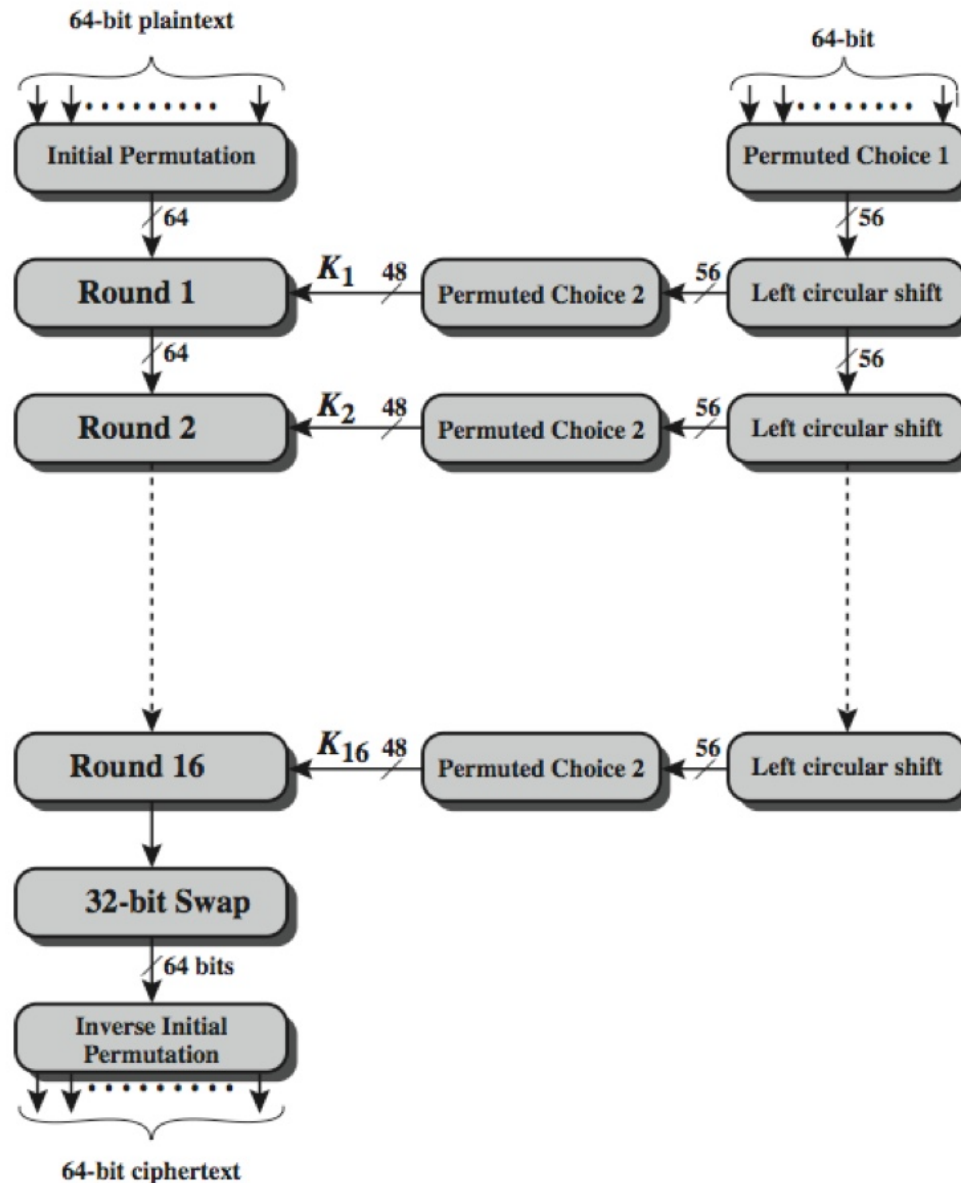
The Data Encryption Standard (DES)

- Considered as the most traditionally used block cipher in the world,
- Created in 1977 by the NBS (now NIST) and carried name: FIPS PUB 46,
- It works by transforming 64-bit data (block) to cipher data using a 56-bit key,
- It was commonly used worldwide, but now has become an obsolete based on several security reports.

The DES Encryption General Overview



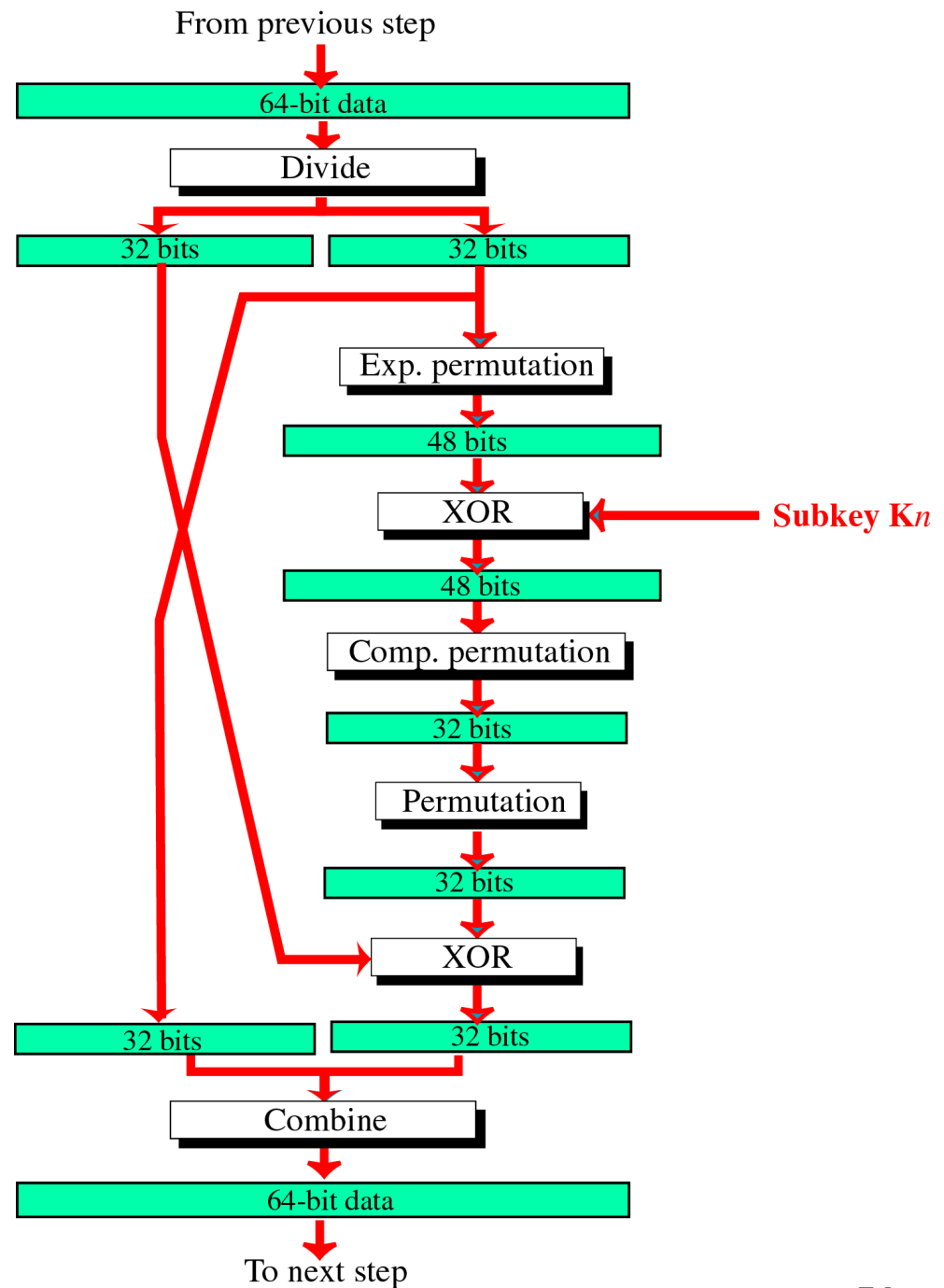
The DES Encryption Detailed Overview



Initial Permutation (IP)

- First step of the data computation,
- IP reorders the input data bits,
- Even bits to LH half, odd bits to RH half,
- Quite regular in structure (easy in h/w),
- Example (in Hex notation):
 - $\text{IP}(675\text{a}6967 \ 5\text{e}5\text{a}6\text{b}5\text{a}) = (\text{ffb}2194\text{d} \ 004\text{df}6\text{fb})$
 - 011001110101101001101001011001110101110010110100110101101011010
 - LH = RH =

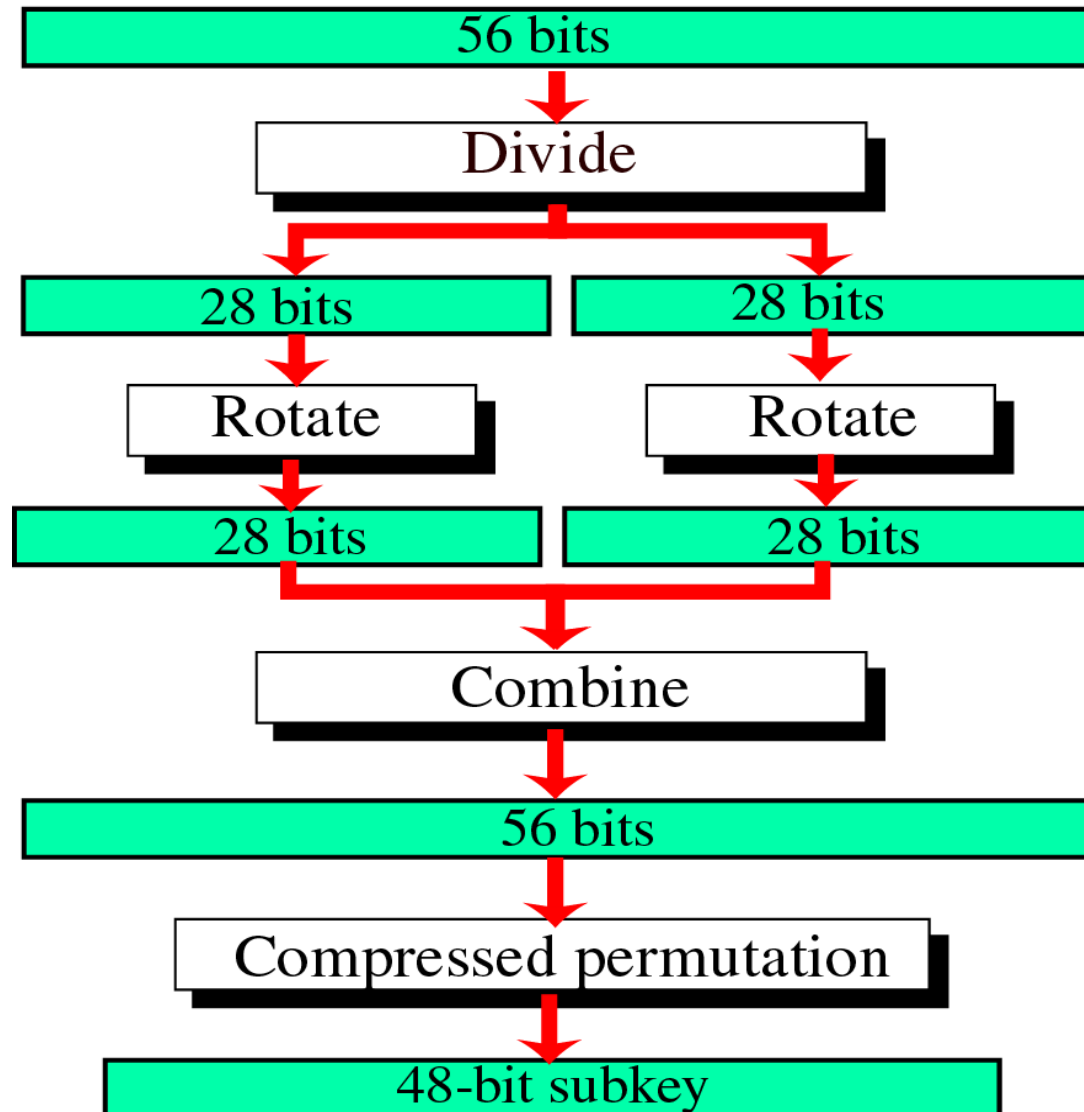
One DES Iteration Block



One DES Iteration Block

- Uses two 32-bit L & R halves,
- Could be described as:
 - $L_i = R_{i-1}$
 - $R_i = L_{i-1} \text{ XOR } F(R_{i-1}, K_i)$
- F takes 32-bit R half and 48-bit subkey:
 - expands R to 48-bits using perm E,
 - adds to subkey using XOR,
 - passes through 8 S-boxes to get 32-bit result,
 - finally permutes using 32-bit perm P

Sub-Key Generations in DES



DES Decryption

- **Decrypt must unwind steps of data computation,**
- **Do encryption steps again using subkeys in reverse order (SK16 ... SK1)**
 - IP undoes final FP step of encryption,
 - 1st round with SK16 undoes 16th encrypt round,
 -
 - 16th round with SK1 undoes 1st encrypt round,
 - then final FP undoes initial encryption IP,
 - thus recovering original data value

DES Example

Plaintext: 02468ace eca86420

Key: 0f1571c947d9e859

subkeys are shown as 6-bit in hex format

Round	K _i	L _i	R _i
IP		5a005a00	3cf03c0f
1	1e030f03080d2930	3cf03c0f	bad22845
2	0a31293432242318	bad22845	99e9b723
3	23072318201d0c1d	99e9b723	Obae3b9e
4	05261d3824311a20	Obae3b9e	42415649
5	3325340136002c25	42415649	18b3fa41
6	123a2d0d04262a1c	18b3fa41	9616fe23
7	021f120b1c130611	9616fe23	67117cf2
8	1c10372a2832002b	67117cf2	c11bfc09
9	04292a380c341f03	c11bfe09	887fbc6c
10	2703212607280403	887fbc6c	600f7e8b
11	2826390c31261504	600f7e8b	f596506e
12	12071c241a0a0f08	f596506e	730538b8
13	300935393c0d100b	738538b8	c6a62c4e
14	311e09231321182a	c6a62c4e	56b0bd75
15	2830300227072528	56b0bd75	75e8fd8f
16	2921080b13143025	75e8fd8f	25896490
IP-1		da02ce3a	89ecac3b

The Avalanche Effect in DES

fourth bit of the
plaintext is
changed

the effect at
the end of first
round

Round		δ
	02468aceeca86420 12468aceeca86420	1
1	3cf03c0fbad22845 3cf03c0fbad32845	1
2	bad2284599e9b723 bad3284539a9b7a3	5
3	99e9b7230bae3b9e 39a9b7a3171cb8b3	18
4	0bae3b9e42415649 171cb8b3ccaca55e	34
5	4241564918b3fa41 ccaca55ed16c3653	37
6	18b3fa419616fe23 d16c3653cf402c68	33
7	9616fe2367117cf2 cf402c682b2cefbcb	32
8	67117cf2c11bfc09 2b2cefbcb99f91153	33

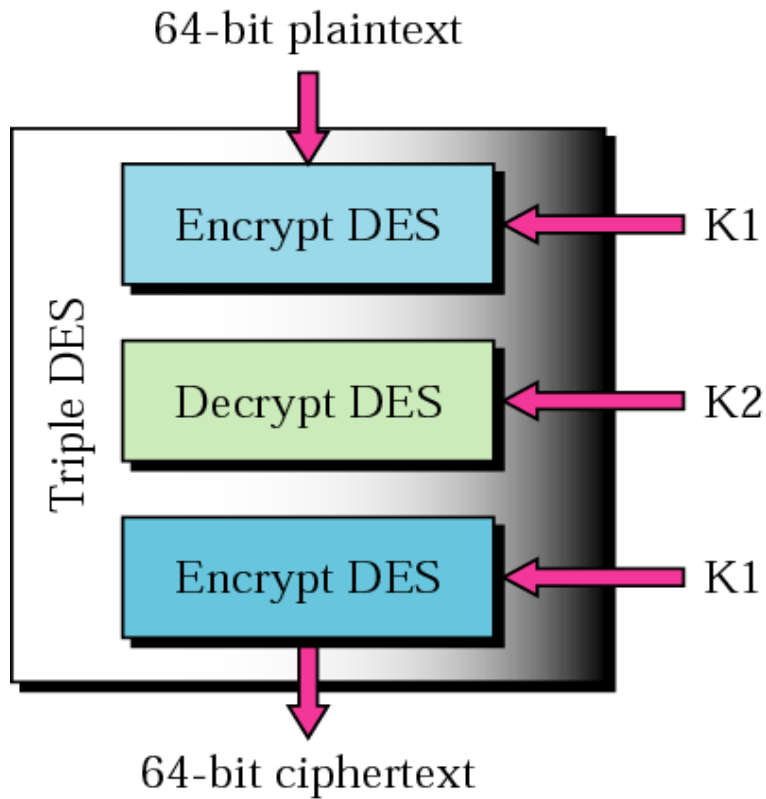
Round		δ
9	c11bfc09887fbc6c 99f911532eed7d94	32
10	887fbc6c600f7e8b 2eed7d94d0f23094	34
11	600f7e8bf596506e d0f23094455da9c4	37
12	f596506e738538b8 455da9c47f6e3cf3	31
13	738538b8c6a62c4e 7f6e3cf34bc1a8d9	29
14	c6a62c4e56b0bd75 4bc1a8d91e07d409	33
15	56b0bd7575e8fd8f 1e07d4091ce2e6dc	31
16	75e8fd8f25896490 1ce2e6dc365e5f59	32
IP ⁻¹	da02ce3a89ecac3b 057cde97d7683f2a	32

the effect at
the end of
reverse Perm

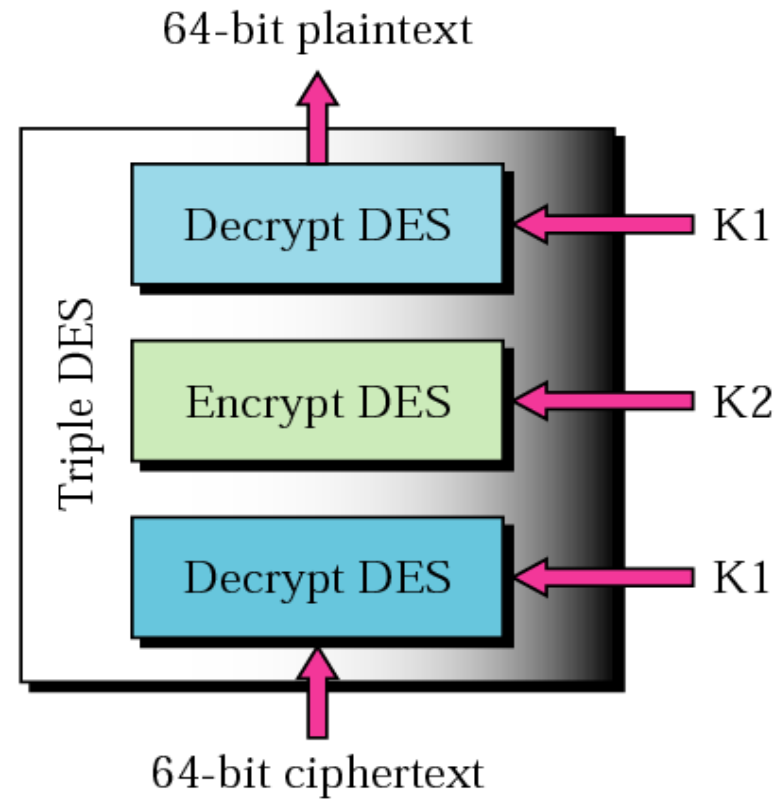
Strength of DES

- 56-bit keys have $2^{56} = 7.2 \times 10^{16}$ values,
- Brute force search looks hard,
- However, recent advances have shown is possible:
 - in 1997 on Internet in a few months,
 - in 1998 on dedicated h/w in a few days,
 - in 1999 above combined in 22hrs!
- Still, attacker must be able to recognize plaintext,
- Must now consider alternatives to DES

Triple DES



a. Encryption triple DES



b. Decryption triple DES

3DES with two keys (112 bits, K1 and K2)
3DES with three keys (168 bits, K1, K2, and K3)

Brute Force Search

- Always possible to simply try every key,
- Most basic attack, proportional to key size,
- Assume either know / recognize plaintext

Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/ μ s	Time required at 10^6 decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu$ s = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu$ s = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu$ s = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu$ s = 5.9×10^{36} years	5.9×10^{30} years

RC4 and RC5

- **RC4 (Rivest Cipher 4) and RC5 are symmetric key cryptographic algorithms developed by Ron Rivest,**
- **RC4 is a fast, byte-oriented stream cipher widely used in legacy protocols (WEP/SSL) but deemed insecure due to vulnerabilities,**
- **RC5 is a flexible, secure block cipher utilizing variable block sizes, rounds, and key lengths**

The IDEA Cryptography Algorithm

- The International Data Encryption Algorithm (IDEA) is a symmetric-key block cipher designed in 1991 by James Massey and Xuejia Lai as a secure, efficient replacement for DES,
- It operates on 64-bit plaintext blocks using a 128-bit key, performing eight identical rounds of encryption followed by an output transformation,
- IDEA is widely used in PGP (Pretty Good Privacy)

The CAST Cryptography Algorithm

- **The CAST algorithm (named for creators Carlisle Adams and Stafford Tavares) is a symmetric-key block cipher used for data encryption,**
- **CAST employs a large 8x32 S-boxes to resist differential/linear cryptanalysis,**
- **It has two variants:**
 - **CAST-128 (64-bit block size, 40-128 bit key), and**
 - **CAST-256 (128-bit block size, up to 256-bit key), widely used in PGP (Pretty Good Privacy) and GPG (GNU Privacy Guard)**

The Blowfish Cryptography Algorithm

- The Blowfish algorithm is a symmetric-key block cipher designed by cryptographer Bruce Schneier in 1993, known for its speed, flexibility, and royalty-free status,
- It operates on fixed-size 64-bit blocks of data and uses a variable-length key, ranging from 32 to 448 bits

The Blowfish Cryptography Algorithm

➤ Key Features:

1. **Symmetric-key:** The same secret key is used for both encryption and decryption,
2. **Block Cipher:** It encrypts data in 64-bit chunks,
3. **Variable Key Length:** The key size can be from 32 to 448 bits, allowing a balance between speed and security,
4. **Structure:** The core of the algorithm is a 16-round structure, which involves splitting the data block into two halves and applying a complex round function repeatedly,
5. **Complex Key Schedule:** A unique feature is its slow and complex key expansion phase, which generates a large number of subkeys (totaling over 4 kilobytes) from the user's secret key before data encryption begins. This slow setup helps protect against brute-force password guessing attacks,
6. **Key-Dependent S-boxes:** The substitution boxes (S-boxes), which are fundamental to the algorithm's security, are generated as part of the key schedule and are unique to each key

